RESUME – Ilker Ozcelik

PERSONAL DATA

Assistant Professor,
ESOGÜ - Meşelik Yerleşkesi
Mühendislik Mimarlık Fakültesi
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February 24, 1981
Eskisehir, Turkey

EDUCATION

Ph.D., Clemson University, Dissertation: ***Denial of Service Attack Detection and Mitigation***, Adviser: Richard R. Brooks, 2015, Electrical Engineering

M.S., Syracuse University, Thesis: ***Voice Activity Detection using Stochastic Resonance***, Adviser: P. Varshney, 2010, Electrical Engineering

B.S., Marmara University, 2005, Electronics and Communication Education

PROFESSIONAL EXPERIENCE

- Eskisehir Osmangazi University, August 2021 – Present, *Assistant Professor*
- Intelligent Systems Security Research Lab, August 2020 - *Director*
- SimCenter, University of Tennessee at Chattanooga, September 2019 -- 2020, *Visiting Scholar and Research Associate*
- Recep Tayyip Erdogan University, 2017 – 2019, *Department Chair of Information Systems Engineering*
- Recep Tayyip Erdogan University, August 2016 – 2021, *Assistant Professor*
- Recep Tayyip Erdogan University, October 2015 – August 2016, *Researcher*
- Clemson University, Fall 2012, Fall 2014 – Spring 2015, *Teaching Assistant*
- Turkish Ministry of National Education, October 2005 – August 2007, *Computer Teacher*
- Cebeci Teletex, May 2005 – October 2005, *Programmer*
- DEMA Relay Industry, July 2003, *Intern*

ACADEMIC AND PROFESSIONAL INTERESTS

- Network Security
- Network Traffic Analysis
- Software Defined Networking
- Adversarial Attacks
- Blockchain
- Cryptographic Functions & Applications

MEMBERSHIPS

- ***Senior Member***, Institute of Electrical and Electronics Engineers, IEEE (*M09, SM25*)
- ***Member***, Association for Computing Machinery, ACM (2014 - 2023)
- ***Member***, Bilgi Güvenliği Derneği (Information Security Association in Turkey), BGD (2016 - 2021)

PROFESSIONAL ACTIVITIES

Projects

- ***Tübitak 3501*** – Scalable Edge Authentication Framework, PI (2023 - 2026)
- ***Tübitak 4004*** – Coding Golden Age, Consultant (2018)
- ***US Department of State (DoS)*** - Internet Freedom for West Africa Project, Research Assistant (2014-2015)
- ***US NSF EAGER-GENI*** - Experiments on Network Security and Traffic Analysis Project, Research Assistant (2010-2012)

Advising

- Elif Değirmenci - Adversarial attack detection on internet of things networks, Co-Advisor, (2024) PhD
- Bilal Alagha - A blockchain-based efficient audit trail framework, Advisor (2024) MSc
- Aybüke KILIÇ - Advanced Adversarial Attack Architecture for Machine Learning Models, Advisor (2025) MSc

Seminars and Working Groups

- Council of Higher Education in Turkey Cyber Security Working Group, Member, (May 2017)
- International Workshop Series on Digital Transformation and Artificial Intelligence, Invited speaker, Istanbul - TÜRKİYE, 2017
- Information Security Association, Information and Cyber Security National Science and Advisory Board, Member, (October 2017 - Present)
- Cyber Security Awareness Workshop, Organiser, Rize - TÜRKİYE, March 2016
- 6th International Certificate Program on: International Cyber Security Workshop Certificate Program, Invited speaker, TÜRKİYE, 2016
- Cyber Security Awareness Workshop, Organiser, Rize - TÜRKİYE, December 2015

Reviews

- IEEE - Transactions on Information Forensics & Security, Reviewer (2019 - Present)
- Elsevier - Computers & Security (COSE), Reviewer (2017 - Present)
- Elsevier - Computer Networks (COMNET), Reviewer (2016 - Present)
- Uluslararası Bilgi Güvenliği Mühendisliği Dergisi (UBGMD), Reviewer (2016 - Present)
- International Journal of Information Security Science (IJISS), Reviewer (2016 - Present)
- Transactions on Dependable and Secure Computing (TDSC), Reviewer (2014)

Other Academic Activities

- International Journal of Information Security Science, *Field Editor* (2024 -)
- Eskişehir Osmangazi Üniversitesi Mühendislik ve Mimarlık Fakültesi Dergisi, *Field Editor* (2021 -)
- Turkish Journal of Electrical Engineering & Computer Sciences, *Guest Editor* (2021)
- Recep Tayyip Erdoğan Üniversitesi Fen ve Mühendislik Bilimleri Dergisi, *Assistant Editor* (2020)

- ISC TURKEY 2020 - Anatolian Crypt Conference, *Security Solutions Track Chair* (2020)
- International Symposium on Networks, Computers and Communications (ISNCC 2019), *Network Function Virtualization and Software Defined Networks Track Chair* (2019)
- Safe Internet for Teens, Sinav Koleji in Ankara, *Presenter* (2019)
- 6th International Symposium on Digital Forensic and Security (ISDFS 2018), *Scientific Committee Member (2018)*
- Second International on Digital Transformation and Artificial Intelligence, *Scientific Committee Member (2017)*
- 3rd International Cyber Warfare and Security Conference, *Attendee (2017)*
- Cyber Security Awareness Workshop, *Organiser,* Rize - TÜRKİYE, March 2016
- DFG/GENI 1st Doctoral Consortium, San Jose, PR, *Attendee (2011)*
- NetFPGA Summer Camp 2011, Stanford University, Stanford, CA, *Attendee (2011)*

Other Industry Collaborations

- ***SAGA Yapay Zeka*** – Cyber Twin (2025 -) Alt Yüklenici PI
- ***DorukNet*** – Datacenter Monitoring and Anomaly Detection System (2024 -), PI
- ***DorukNet*** – Detectiong DDoS Attack Sources in Autonomous Systems (2023), PI
- ***ASELSAN*** – Cyber Security Research and Development Project (2023), Researcher
- ***Enerji 5G*** (2023), Researcher
- ***Fenix*** - Yerli Enerji Yerli Teknoloji Ar-Ge Projesi (2022), Researcher
- ***R.I.S.E. Technology*** TEYDEB 1501 R&D Project, *Project Advisor*

PUBLICATIONS

Books and Monographs

- Bilal Alagha and **İlker Özçelik**. " Using Blockchain Technology for Audit Trail." Digital Strategy and Governance in Transformative Technologies, pp. xxx-xxx. CRC Press, 2025.
- Sağıroğlu, Süreyya Atasever-**İlker Özçelik**-Şeref. "Dağıtık Hizmet Engelleme/Aksattırma Saldırıları (DDoS) ve Karşı Önlemler." Siber Güvenlik ve Savunma Kitap Serisi 6: Siber Güvenlik Ontolojisi, Tehditler ve Çözümler 6: 221. Nobel, 2022
- Bhat, Naazira B., Dulip Madurasinghe, **Ilker Ozcelik**, Richard R. Brooks, Ganesh Kumar Venayagamoorthy, and Anthony Skjellum. "Evaluation and Design of Performable Distributed Systems." In Handbook of Advanced Performability Engineering, pp. 211-227. Springer, Cham.
- **Özçelik, İlker,** and Richard Brooks. Distributed denial of service attacks: Real-world detection and mitigation. CRC Press, 2020.
- Brooks, R. R., Griffin, C., Friedlander, D. S., Koch, J. D., & **Özçelik, I**. Target Tracking with Self-Organizing Distributed Sensors in *Distributed Sensor Networks: Image and Sensor Signal Processing*, 131-189. Chapman & Hall/CRC Press, 2012.

Refereed Journal Publications

- Çekiş, İsmet Kaan, Buğra Ayrancı, Fezayim Numan Salman, and **İlker Özçelik**. "IP Spoofing Detection Using Deep Learning." *Applied Sciences* 15, no. 17 (2025): 9508.
- B. Alagha and **I. Ozcelik**, "BEATS: Practical Audit Trail in Blockchain Systems," in IEEE Access, vol. 13, pp. 109657-109669, 2025, doi: 10.1109/ACCESS.2025.3582722.
- Değirmenci, Elif, **Ilker Özçelik**, and Ahmet Yazıcı. "Adversarial Attack Detection Approach for Intrusion Detection Systems." IEEE Access (2024).
- Çekiş, İsmet Kaan, Armağan Toros, Nimet Apaydın, and **İlker Ozcelık**. "Performance comparison of ECC libraries for IoT devices." Eskişehir Technical University Journal of

Science and Technology A-Applied Sciences and Engineering 25, no. 2 (2024): 278-288.

- Alagha, Bilal, **İlker Özçelik**, and Emrah Atılgan. "Smart Agriculture Blockchain Applications." Selcuk Journal of Agriculture and Food Sciences 38, no. 3 (2024): 578-605.
- Ertem, Mehmet, and **İlker Ozcelik**. "Siber ağların risk analizi: Saldırı-savunma ağaçlarıyla temellendirilmiş niceliksel bir yaklaşım." Journal of Innovative Engineering and Natural Science 4, no. 1 (2024): 113-125.
- Değirmenci, Elif, Yunus Sabri Kırca, **İlker Özçelik**, and Ahmet Yazıcı. "ROSIDS23: Network intrusion detection dataset for robot operating system." Data in Brief 51 (2023): 109739.
- Oakley, Jonathan, Carl Worley, Lu Yu, Richard R. Brooks, **İlker. Özçelik**, Anthony Skjellum, and Jihad S. Obeid. "Scrybe: A Secure Audit Trail for Clinical Trial Data Fusion." Digital threats: research and practice 4, no. 2 (2023): 1-20.
- **Ozcelik, Ilker**, Sai Medury, Justin Broaddus, and Anthony Skjellum. "An overview of cryptographic accumulators." arXiv preprint arXiv:2103.04330 (2021).
- Brooks, R. R., Lu Yu, **Ilker Ozcelik**, Jon Oakley, and Nathan Tusing. "Distributed denial of service (DDoS): a history."IEEE Annals of the History of Computing 44, no. 2 (2021): 44-54.
- Atasever, S., **Ozcelik, I.**, & Sagiroglu, S. (2019). Siber Teror ve DDoS. SDÜ Fen Edebiyat Fakültesi Fen Bilimler Enstitüsü Dergisi, 23(1), 227-233.
- Fu, Y., Yu, L., Hambolu, O., **Ozcelik, I.**, Husain, B., Sun, J., ... & Brooks, R. R. (2017). Stealthy Domain Generation Algorithms. *IEEE Transactions on Information Forensics and Security*, *12*(6), 1430-1443.
- **Özçelik, İ**., & Brooks, R. R. (2015). Deceiving entropy-based DoS detection. *Computers & Security*, *48*, 234-245.

<u>Conference Proceedings (Reviewed)</u>

- Toros, Armağan, İsmet Kaan Çekiş, Kübra Korkmaz, and **İlker Özçelik.** "Performance Evaluation of ECC Based Cryptographic Accumulator on Resource-Constrained Platform." In 2024 Innovations in Intelligent Systems and Applications Conference (ASYU), pp. 1-6. IEEE, 2024.
- Degirmenci, Elif, **Ilker Ozcelik**, and Ahmet Yazici. "Effects of Un targeted Adversarial Attacks on Deep Learning Methods." In 2022 15th International Conference on Information Security and Cryptography (ISCTURKEY), pp. 8-12. IEEE, 2022.
- Değırmencı, Elif, **İlker Özçelık**, and Ahmet Yazici. "The Effects of Autoencoders on the Robustness of Deep Learning Models." In 2022 30th Signal Processing and Communications Applications Conference (SIU), pp. 1-4. IEEE, 2022.
- Atilgan, Emrah**, Ilker Ozcelik**, and Esra N. Yolacan. "MQTT Security at a Glance." In 2021 International Conference on Information Security and Cryptology (ISCTURKEY), pp. 138-142. IEEE, 2021.
- **Ozcelik, Ilker**. "Capen: Cryptographic accumulator based privacy preserving exposure notification." In 2021 9th International Symposium on Digital Forensics and Security (ISDFS), pp. 1-6. IEEE, 2021.
- **Ozcelik, Ilker,** and Anthony Skjellum. "Cryptorevocate: A cryptographic accumulator based distributed certificate revocation list." In 2021 IEEE 11th Annual Computing and Communication Workshop and Conference (CCWC), pp. 0865-0872. IEEE, 2021.
- **Ozcelik, Ilker,** Sai Medury, Justin Broaddus and Anthony Skjellum. "An Overview of Cryptographic Accumulators." In, The International Conference on Information Systems Security and Privacy (ICISSP 2021) .
- Atasever, Süreyya, **İlker Özçelık**, and Şeref Sağiroğlu. "An Overview of Machine Learning Based Approaches in DDoS Detection." In 2020 28th Signal Processing and Communications

Applications Conference (SIU), pp. 1-4. IEEE, 2020.

- F. Kandah, **I. Ozcelik**, and B. Huber, "MARS: Machine learning based Adaptable and Robust Network Management for Software-defined Networks" In *,10th Annual Computing and Communication Workshop and Conference (CCWC), 2020*. IEEE.
- **Ozcelik**, S. Akleylek, and I. Ozcelik, "TRCyberLab: An Infrastructure for Future Internet and Security Studies" In *International Symposium on Digital Forensic and Security (ISDFS), 2018*. IEEE.
- **Özçelik, İ**., & Brooks, R. R. (2016, April). Cusum-entropy: an efficient method for DDoS attack detection. In *Smart Grid Congress and Fair (ICSG), 2016 4th International Istanbul* (pp. 1-5). IEEE.
- **Ozcelik**, Y. Fu, and R. Brooks, "Dos detection is easier now," Second GENI Research and Educational Experiment Workshop, (March 2013).
- **Ozcelik** and R. Brooks, "Operational system testing for design in security," Proceedings of the 8th Annual Workshop on Cyber Security and Information Intelligence Research, Oak Ridge, TN, (January 2013).
- **Ozcelik** and R. R. Brooks, "Performance Analysis of DDoS Detection Methods on Real Network," The First GENI Research and Educational Experiment Workshop, Los Angeles, CA (March 2012).
- **Ozcelik**, and R. R. Brooks, "Security Experimentation Using Operational Systems," "Proceedings of the 7thAnnual Workshop on Cyber Security and Information Intelligence Research, Oak Ridge, TN (October 2011).
- H. Bhanu, J. Schwier, R. Craven, **I. Ozcelik**, C. Griffin, and R. R. Brooks, "Noise tolerant symbolic learning of markov models of tunneled protocols," in Wireless Communications and Mobile Computing Conference (IWCMC), 2011 7th International, pp. 1310–1314, IEEE, 2011.

Conference Proceedings (Unreviewed)

- **Ozcelik**, R. R. Brooks, Deceiving entropy-based DoS detection," Signal Processing, Sensor/Information Fusion, and Target Recognition XXIII. SPIE, Baltimore, MD, (May 2014).

HONORS AND AWARDS

- International Postdoctoral Research Fellowship, The Scientific and Technological Research Council of Turkey (2018)
- Best Idea Award by both peers and faculty at DFG/GENI 1st Doctoral Consortium, San Jose, PR (Group Project) (March 2011)
- Full Scholarship, Ministry of National Education of the Republic of Turkey. (2007 – 2015)

TEACHING

Courses Taught

- 503002512, Applied Network Security, Eskisehir Osmangazi Üniversitesi, Spring 22, Fall 24
- 152118640, Engineering Research on Cyber Security, Eskisehir Osmangazi Üniversitesi, Fall, Spring 23, 24
- 152118514, Network Security, Eskisehir Osmangazi Üniversitesi, Fall 22, Spring 23, 24
- 152116027, Computer Networks, Eskisehir Osmangazi Üniversitesi, Spring 23, 24
- 152113021, Data Structures, Eskisehir Osmangazi Üniversitesi, Fall 22, 23, 24
- EEE 206, Signals and Systems, Recep Tayyip Erdoğan Üniversitesi, *Spring 19*
- EEE 204, Probability, Recep Tayyip Erdoğan Üniversitesi, *Spring 17,18,19*

- INS 124, Computer Programming, Recep Tayyip Erdoğan Üniversitesi, *Spring 17,18*
- ENF 161, Fundamental Information Technologies, Recep Tayyip Erdoğan Üniversitesi, *Fall 16,17,18*
- EEE 101, Introduction to Programming, Recep Tayyip Erdoğan Üniversitesi, *Fall 16, 17,18*